17

Oracle on Oracle

Objectives

After completing this lesson, you should be able to:

- Prepare your Oracle Linux server for Oracle Database installation
- Use Oracle Pre-Install RPM
- Create Oracle software user and group accounts
- Set kernel parameters for Oracle Database
- Set Oracle database shell limits
- Configure HugePages
- Configure Oracle Database Smart Flash Cache (DBSFC)
- Describe Oracle ASM, ASM Library Driver (ASMLib), and ASM Filter Driver (ASMFD)

Oracle Pre-Install RPM

Oracle Database Pre-Install RPM for Oracle Linux:

- Completes most pre-installation configuration tasks
- Downloads and installs various software packages and specific versions needed for database installation
- Creates the user `oracle` and the groups `oinstall` and `dba`
- Updates kernel parameters
  - These reside in `/etc/sysctl.d/99-oracle-database-server-18c-preinstall-sysctl.conf`
- Sets hard and soft shell resource limits in the `/etc/security/limits.d` directory
- Sets `numa=off` in the kernel boot parameters for x86_64 machines
- Sets `transparent_hugepage=never` in the kernel boot parameters

ORACLE

Oracle Software User Accounts

- The Oracle database software owner:
  - Is commonly named `oracle`
  - Runs the OUI and has full privileges to install, uninstall, and patch Oracle software
  - Cannot be `root`
- The owner of the `httpd` process is:
  - A low-privileged OS user
  - Usually provided by the `nobody` user
- Database operations require a few more users:
  - Members of `OSOPER` group can start, stop, back up, and recover the database.
  - Members of the `OSDBA` group have `OSOPER` privileges, can create and drop a database, and create other `OSDBA` members.

Oracle Automatic Storage Management (ASM) Groups

- The `OSASM` group for Oracle ASM Administration:
  – This is commonly named `asmadmin`.
  – Create this group as a separate group if you want to have separate administration privileges groups for Oracle ASM and Oracle Database administrators.
- The `OSDBA` group for Oracle ASM:
  – This is commonly named `asmdba`.
  – The Oracle Grid Infrastructure software owner (typically, `grid`) must be a member.
  – Membership in this group enables access to the files managed by Oracle ASM.
- The `OSOPER` group for Oracle ASM:
  – This is an optional group.
  – This is commonly named `asmoper`.
  – Create this group if you want a separate group of operating system users to have a limited set of Oracle instance administrative privileges.

ORACLE

For example:

```
# usermod -aG asmadmin oracle
# usermod -aG asmdba oracle
```

System Resource Tuning

- An Oracle database instance requires certain system resources.
- Shared memory must be adjusted for database use.
- Shared memory system uses semaphores, which must be adjusted.
- Each dedicated server process requires a network port.
- Larger network buffers are recommended.
- The maximum number of open files per process must be increased.
- Shell limit settings must be increased.

Linux Shared Memory

- Three shared memory-related kernel parameters:
  - `SHMMNI`: The maximum number of system-wide shared memory segments
  - `SHMMAX`: The maximum size of each segment
  - `SHMALL`: The maximum number of shared memory pages system wide
- For Oracle database, set `SHMMAX` >= the largest SGA.
- Set shared memory kernel parameters in: `/etc/sysctl.d/97-oracle-database-sysctl.conf`
- Parameters are viewable in:

```
# ls /proc/sys/kernel/sh*
shmall      shmmax      shmmni
```

Semaphores

- Semaphores are a method of controlling access to critical resources.
- The Oracle instance uses semaphores to control access to shared memory.
- Semaphores are allocated based on the `PROCESSES` initialization parameter.
- All four semaphore parameters are set by a single `kernel.sem` parameter in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:
  - `semmsl`: Maximum number of semaphores per set
  - `semmns`: Total number of semaphores in the system
  - `semopm`: Maximum number of operations per `semop` call
  - `semmni`: Maximum number of semaphore sets

Minimum configurations:

- For `semmsl`: 250
  - or the largest `PROCESSES` parameter of an Oracle database plus 50
- For `semmns`: 32000
  - or sum of the `PROCESSES` parameters for each Oracle database, and adding an additional 25 to 50 for each database
- For `semopm`: 100
- For `semmni`: 128

Setting in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:

`kernel.sem = 250 32000 100 128`

Network Tuning

- Socket parameters:
  - An IP port is assigned to a server process when it starts.
  - An IP port is used to communicate with the user process.
  - The default range is 32768 through 61000. Change it if not adequate. Example:

```
# cat /proc/sys/net/ipv4/ip_local_port_range
9000    65500
```

- TCP/IP window size parameters:
  - Define read (`rmem`) and write (`wmem`) window sizes.
  - Set the default and maximum memory allocated for the network send and receive buffers.

```
# ls /proc/sys/net/core/[rw]mem*
rmem_default        wmem_default
rmem_max            wmem_max
```

ORACLE

Network Tuning - Recommendation

- Change the socket receive buffer:

```
#  sysctl -w net.core.rmem_default=262144
```

- Change the socket send buffer:

```
#  sysctl -w net.core.wmem_default=262144
```

- Change the maximum socket receive buffer size:

```
#  sysctl -w net.core.rmem_max=262144
```

- Change the maximum socket send buffer size

```
#  sysctl -w net.core.wmem_default=262144
```

Setting the File Handles Parameter

- The File Handles parameter (`fs.file-max`) determines the maximum number of file handles that the Linux kernel allocates.
- The Oracle database background processes open all data files, logs, and other supporting files.
- The parameter must be high enough to include all the data files within your database and all supporting files.
- Set the kernel parameter in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:

  ```
  fs.file-max = 6815744
  ```

- View the setting in:

```
# cat /proc/sys/fs/file-max
6815744
```

Asynchronous I/O (AIO)

- AIO is the kernel subsystem used to ensure that Oracle databases run properly on Linux.
- AIO allows a process to initiate several I/O operations without having to block or wait for any to complete.
- The process can retrieve the results of the I/O later.
- Set the maximum number of allowable concurrent requests kernel parameter in `/etc/sysctl.d/97-oracle-database-sysctl.conf`:

  ```
  fs.aio-max-nr = 1048576
  ```

- View the setting in:

```
# cat /proc/sys/fs/aio-max-nr
1048576
```

Oracle-Related Shell Limits

- Three shell limits must be set for the `oracle` user:
  - `nofile`: Number of open file descriptors
  - `nproc`: Number of processes available to a single user
  - `stack`: Size of the stack segment of the process
- Soft limit versus hard limit
  - A hard limit can be changed only by `root`.
  - A soft limit can be changed by the user, up to the value of the hard limit.
- Define limits in the `/etc/security/limits.conf` file.
- Edit the `/etc/pam.d/login` file.
- A user can change a soft limit by using the `ulimit` command, for example:

```
$ ulimit -Sn 50
```

ORACLE

HugePages

- HugePages:
  - Allow larger pages to manage memory
  - Are crucial for faster Oracle database performance
  - Are useful in both 32- and 64-bit configurations
  - Are integrated into the Linux kernel with release 2.6
  - Have been back-ported to some 2.4 kernels (2.4.21), but are implemented differently
  - Decrease page table overhead
  - Provide faster overall memory performance
  - Must be reserved during system startup
  - Are not swappable—there is no page-in/page-out overhead
- HugePage sizes vary from 2 MB to 256 MB, based on the kernel version and the hardware architecture.

Configuring HugePages

- Guidelines exist for different OS versions and hardware architectures.
- Configuring HugePages on 64-bit Linux:
  - Set the `memlock` user limit in the appropriate file in `/etc/security/limits.d` slightly smaller than installed RAM.
  - Disable AMM by setting `MEMORY_TARGET` and `MEMORY_MAX_TARGET` to zero.
  - Use the `hugepages_settings.sh` script to calculate the recommended value for the `vm.nr_hugepages` parameter.
  - Edit `/etc/sysctl.d/97-oracle-database-sysctl.conf` and set the `vm.nr_hugepages` parameter.
  - Reboot your system.
- To check: `grep HugePages /proc/meminfo`

Oracle Database Smart Flash Cache (DBSFC)

- DBSFC:
  - Is available for both Oracle Solaris and Oracle Linux customers with the 11g R2, 12c, and 18c databases
  - Allows you to extend the Oracle Buffer Cache in memory (SGA) using secondary flash-based storage
  - Helps with read-only/read-mostly workloads
- When a block gets modified, it is modified in the standard database buffer cache, written to disk and copied over into the flash cache.
- A subsequent read can then be from this fast storage instead of from the originating data files.
- See https://www.oracle.com/technetwork/articles/servers-storage-admin/smart-flash-cache-oracle-perf-361527.html.

Oracle ASM

- For stand-alone or Oracle RAC databases, you must configure ASM on shared disk that is accessible by all cluster nodes.
  - Creating Oracle Clusterware files on shared file systems is deprecated.
- ASM performs the functions of a volume manager and a file system.
- ASM consists of a specialized Oracle instance and a set of disk groups that are managed through the ASM instance.
- A disk group is a set of disk devices that ASM manages.
  - Each disk device can be a partition, a logical volume, a RAID array, or a single disk.
  - ASM spreads data evenly across the disk group to optimize performance and usage.

ORACLE

ASM Library Driver (ASMLib)

- ASMLib simplifies the management of ASM disks.
- ASMLib has three components:
  - `oracleasm-support`: Provides user space shell scripts and is included with the Oracle Linux distribution
  - `oracleasmlib`: Provides the user space library and is installed from Unbreakable Linux Network (ULN)
  - `oracleasm`: Is the kernel driver included in `kernel-uek`
- To configure ASMLib:

```
# oracleasm configure -i
```

- To mark disks as ASM disks:

```
# oracleasm createdisk ASM_DISK_NAME candidate_disk
```

Using ASMLib Commands

Some options for the `oracleasm` program:

- `configure`: Configure the ASMlib driver. Options:
  - `–i` - interactive mode; `–e` - enable ASMLib startup on boot; `–d` - disable ASMLib startup on boot.
- `init`: Load and initialize the ASMLib driver.
- `exit`: Stop the ASMLib driver.
- `createdisk`: Mark a disk device for use with ASM.
- `deletedisk`: Unmark a named disk device.
- `querydisk`: Determine whether a disk device or disk name is being used by the ASMLib.
- `listdisks`: List the disk names of marked disks.
- `scandisks`: Check block devices for ASM disks.

ASM Filter Driver (ASMFD)

- Available on Linux systems starting with Oracle Database 12c Release 1
- Is an alternative to ASMLib
- Prevents accidental corruption or deletion of ASM devices
- Provides persistent paths and permissions for storage devices used with ASM
  – No need to update `udev` files with storage device paths and permissions.
- Cannot be run concurrently with ASMLib
  – If ASMLib is installed, it must be deinstalled before running ASMFD.

ORACLE

Quiz

Which of the following statements is true regarding ASM?

a. ASMLib is required to use ASM.

b. The `oracleasm` kernel driver is included in the Red Hat Compatible Kernel (RHCK).

c. When using ASMLib, for Oracle Universal Installer (OUI) to recognize partitions as Oracle ASM disk candidates, you must mark the disk partitions that Oracle ASM can use.

d. A RAID array cannot be included in an ASM disk group.

Quiz

Which of the following statements is true regarding ASM?

a. ASMLib is required to use ASM.

b. The `oracleasm` kernel driver is included in the Red Hat Compatible Kernel (RHCK).

c. When using ASMLib, for Oracle Universal Installer (OUI) to recognize partitions as Oracle ASM disk candidates, you must mark the disk partitions that Oracle ASM can use.

d. A RAID array cannot be included in an ASM disk group.

Quiz

Which of the following statements is true regarding ASM?

a. ASMLib is required to use ASM.

b. The `oracleasm` kernel driver is included in the Red Hat Compatible Kernel (RHCK).

c. When using ASMLib, for Oracle Universal Installer (OUI) to recognize partitions as Oracle ASM disk candidates, you must mark the disk partitions that Oracle ASM can use.

d. A RAID array cannot be included in an ASM disk group.

Summary

In this lesson, you should have learned how to:

- Prepare your Oracle Linux server for Oracle Database installation
- Use Oracle pre-install RPM
- Create Oracle software user and group accounts
- Set kernel parameters for Oracle Database
- Set Oracle database shell limits
- Configure HugePages
- Configure Oracle Database Smart Flash Cache (DBSFC)
- Describe Oracle ASM, ASM Library Driver (ASMLib), and ASM Filter Driver (ASMFD)

Practices for Lesson 17: Overview

This practice covers the following topics:

- Using `scp` to upload `oracle` packages
- Installing and running Oracle Database Pre-Install
- Preparing disks for ASM use
- Installing and configuring ASMLib
- Reverting changes made to host03